

A0

one generator, views and details of the same vertices

Index

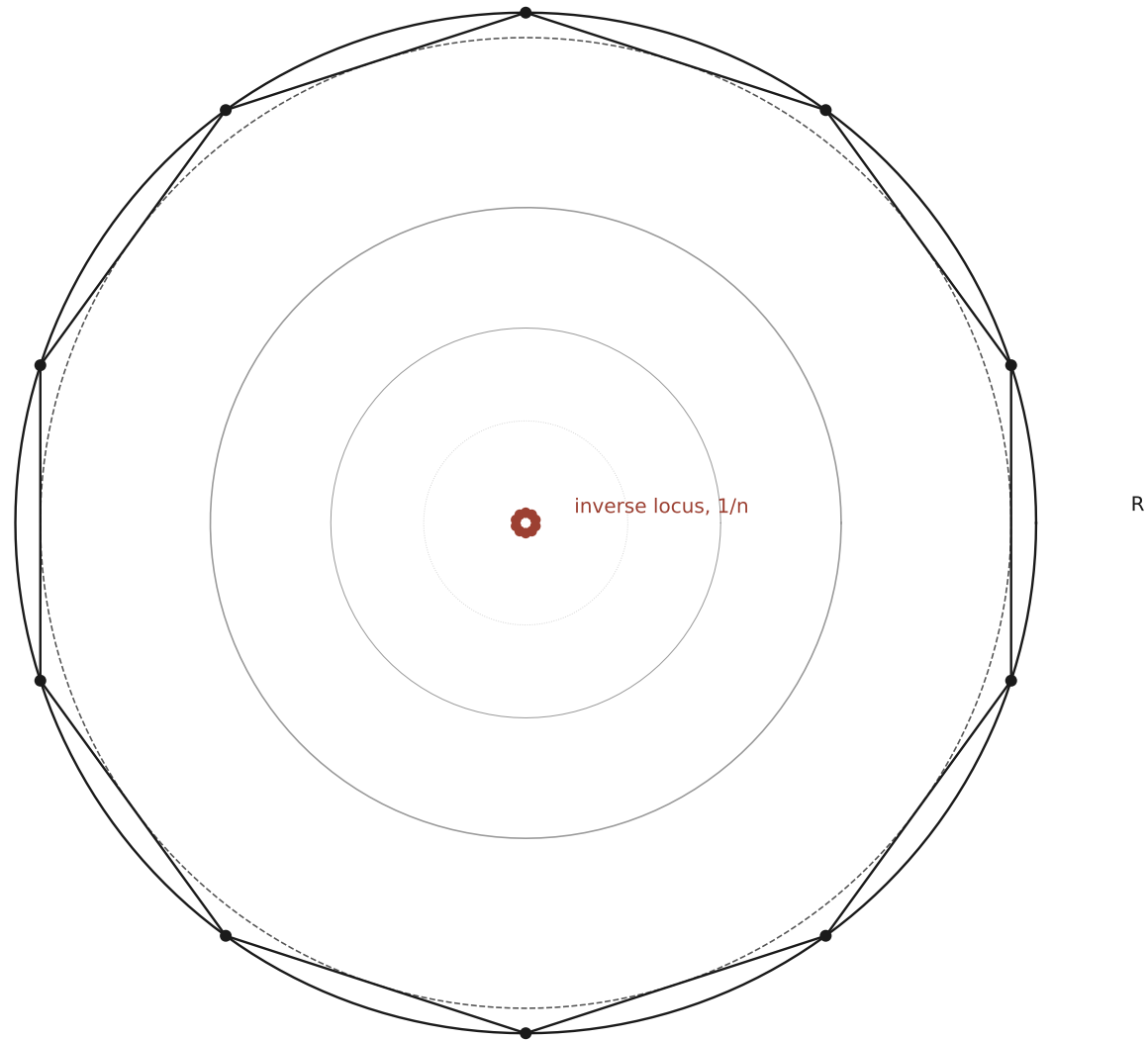
n equally spaced vertices, radius $n/2$. Outer locus in ink.
The dry red is the inverse locus, always: the same vertices at radius $1/n$.
Nothing is drawn that is not a vertex or a ring of this kernel.

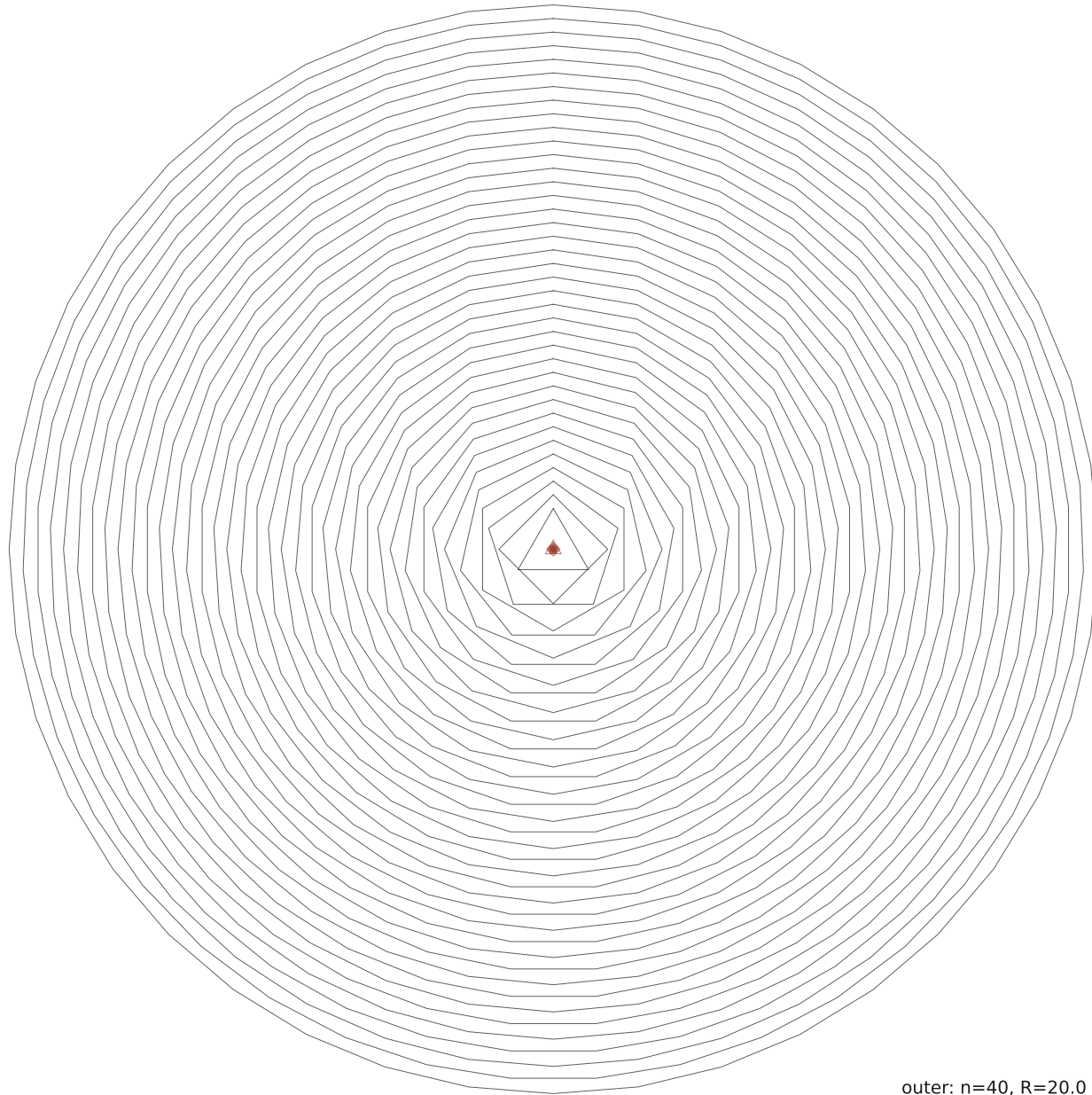
P1 PLAN one n-gon outer, apothem, inverse locus	P2 OVERLAY every n, concentric outer in ink, inverse in red	E1 ELEVATION Archimedes squeeze inscribed / circumscribed on pi	D1 THE RING apothem, B1, B2, inverse vs n
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D2 lock n=1 D3 marriage n=5 D4 vertex structure n=12

SCH1 totient schedule SCH2 reducible density

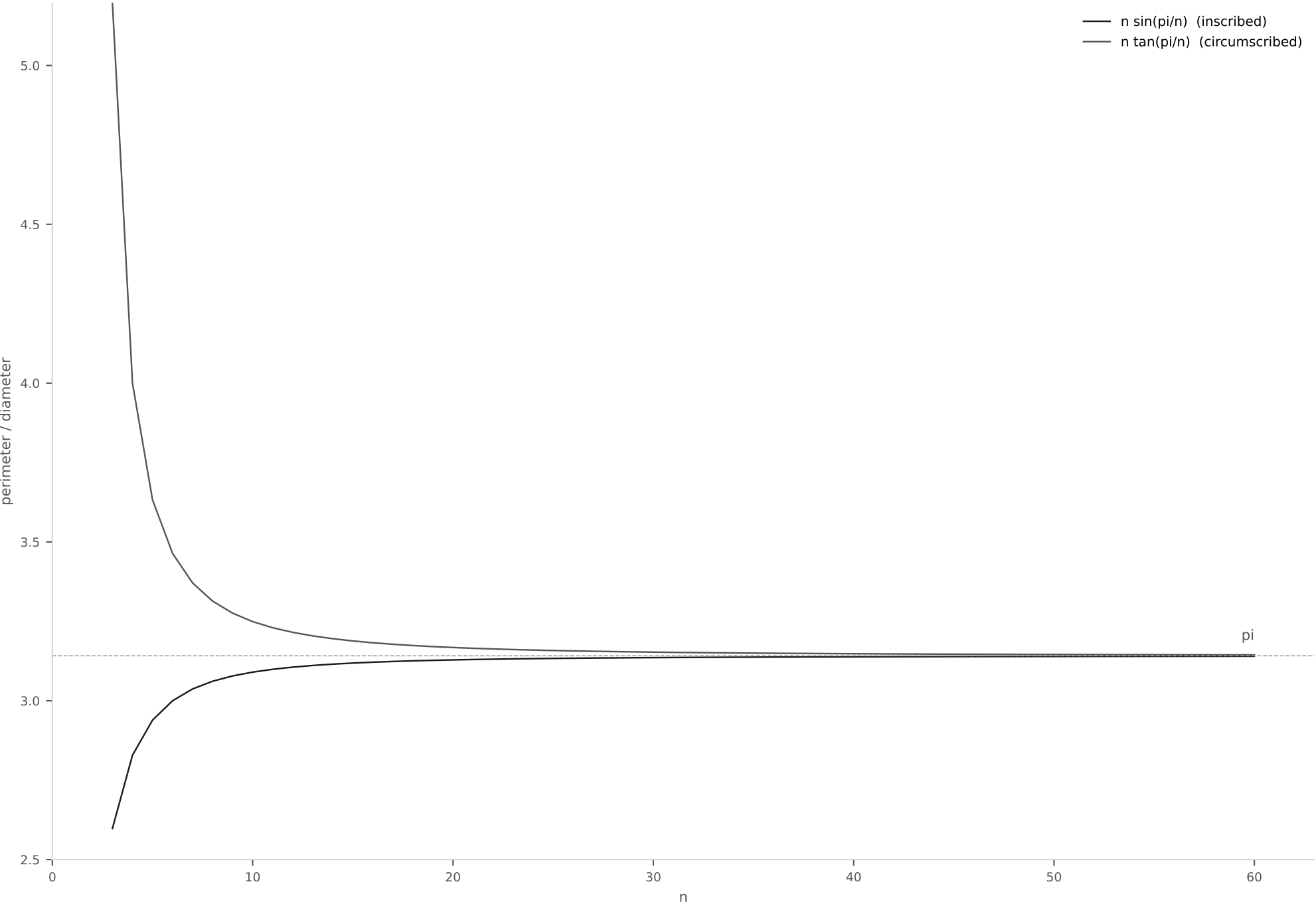
this run: n = 1,60





outer: $n=40$, $R=20.0$

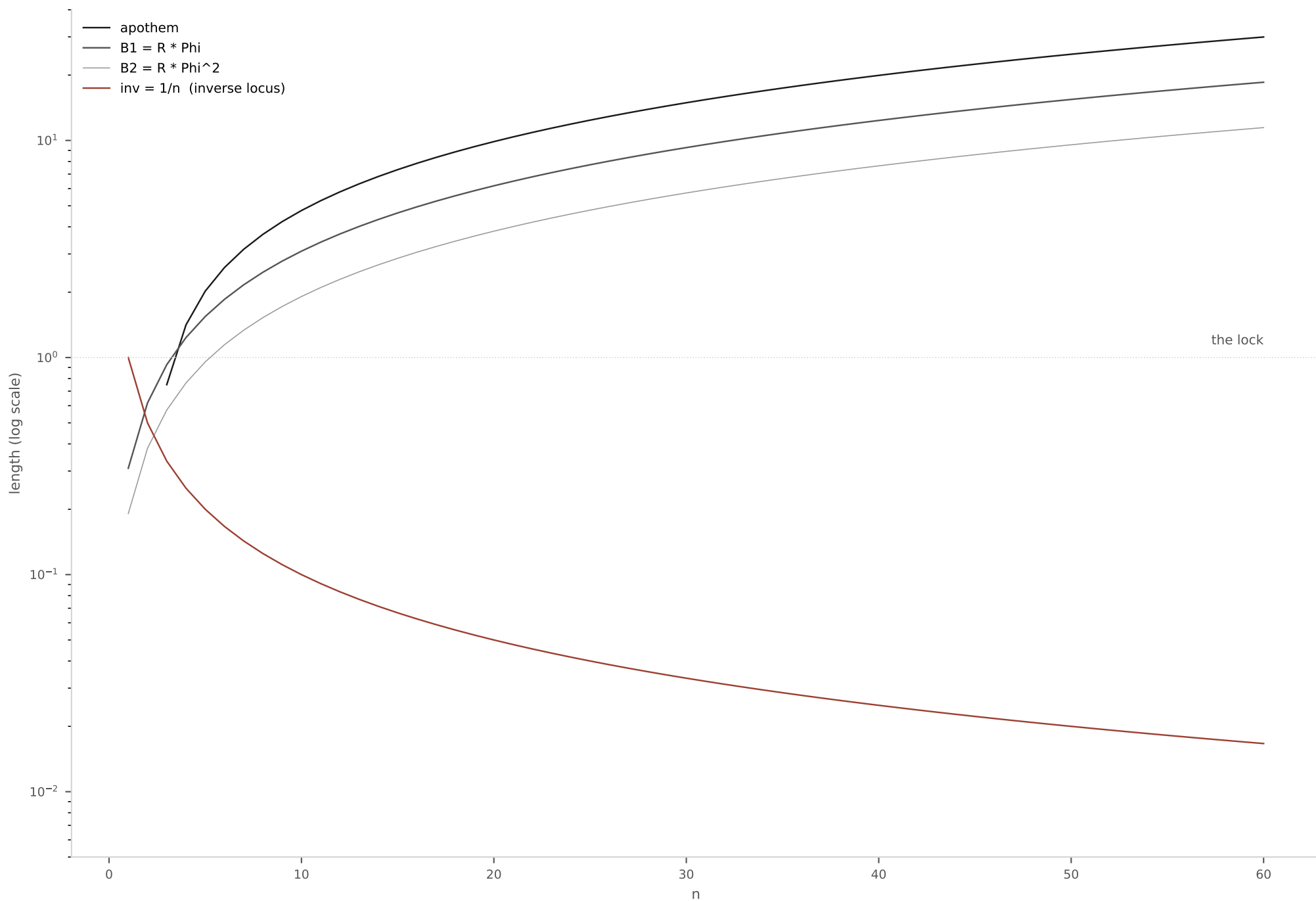
inverse: same n , radius $1/n$



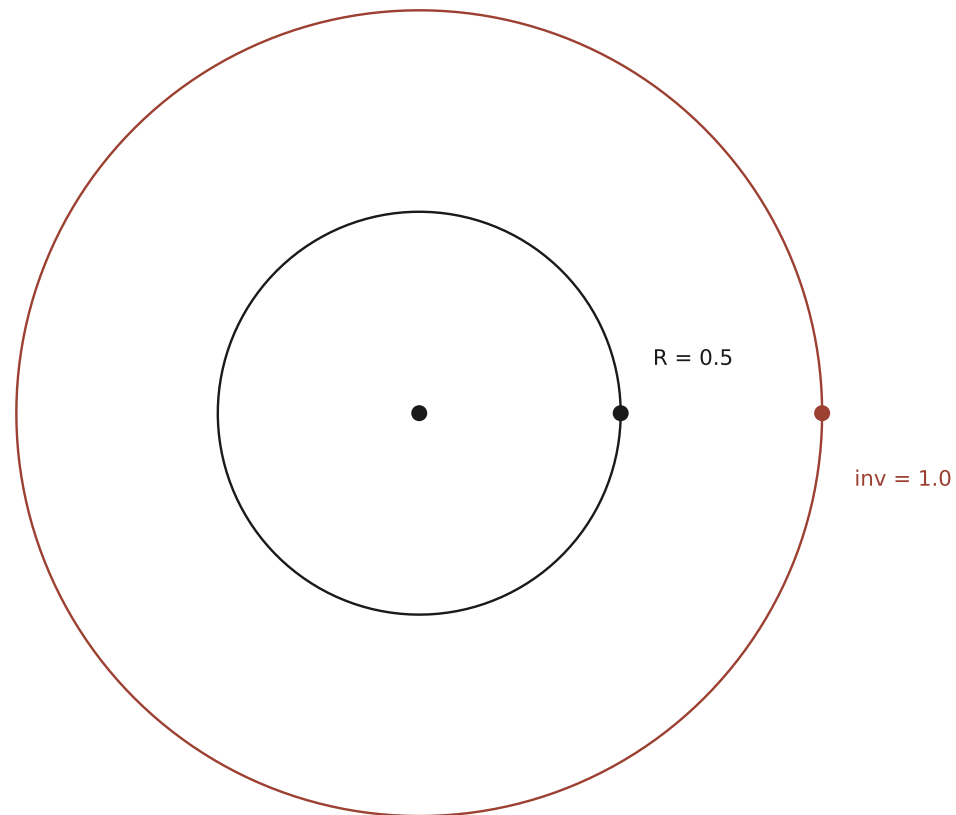
D1

apothem, the two Phi-nests, and the inverse locus, from rings.csv

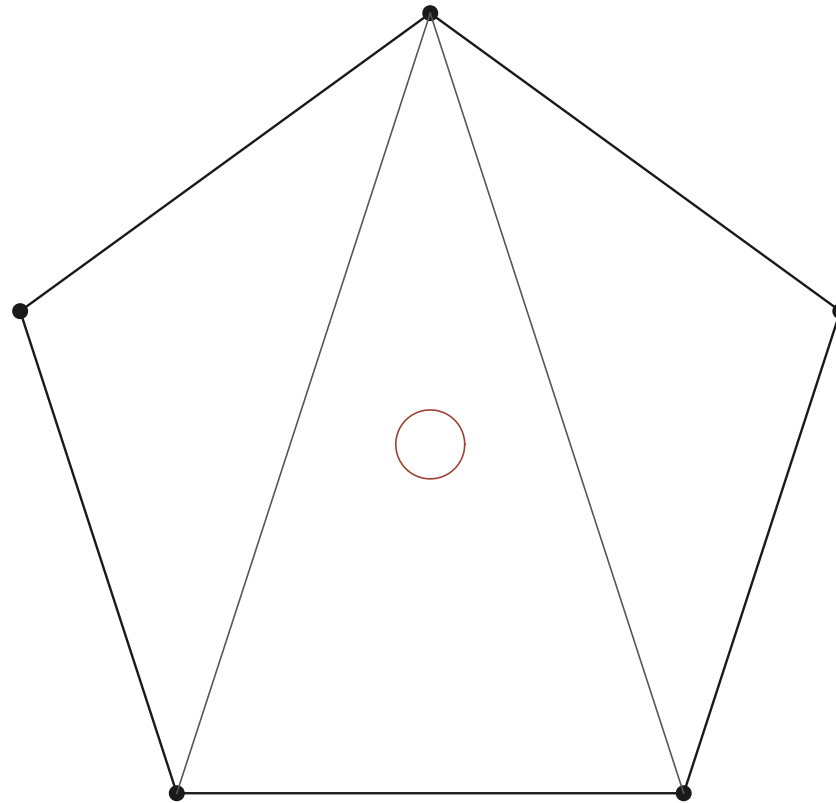
The ring



the outer locus and the inverse locus start at the same point



at $n=1$, the outer diameter is 1 and the inverse locus is also exactly 1 -- the one place the two loci coincide.

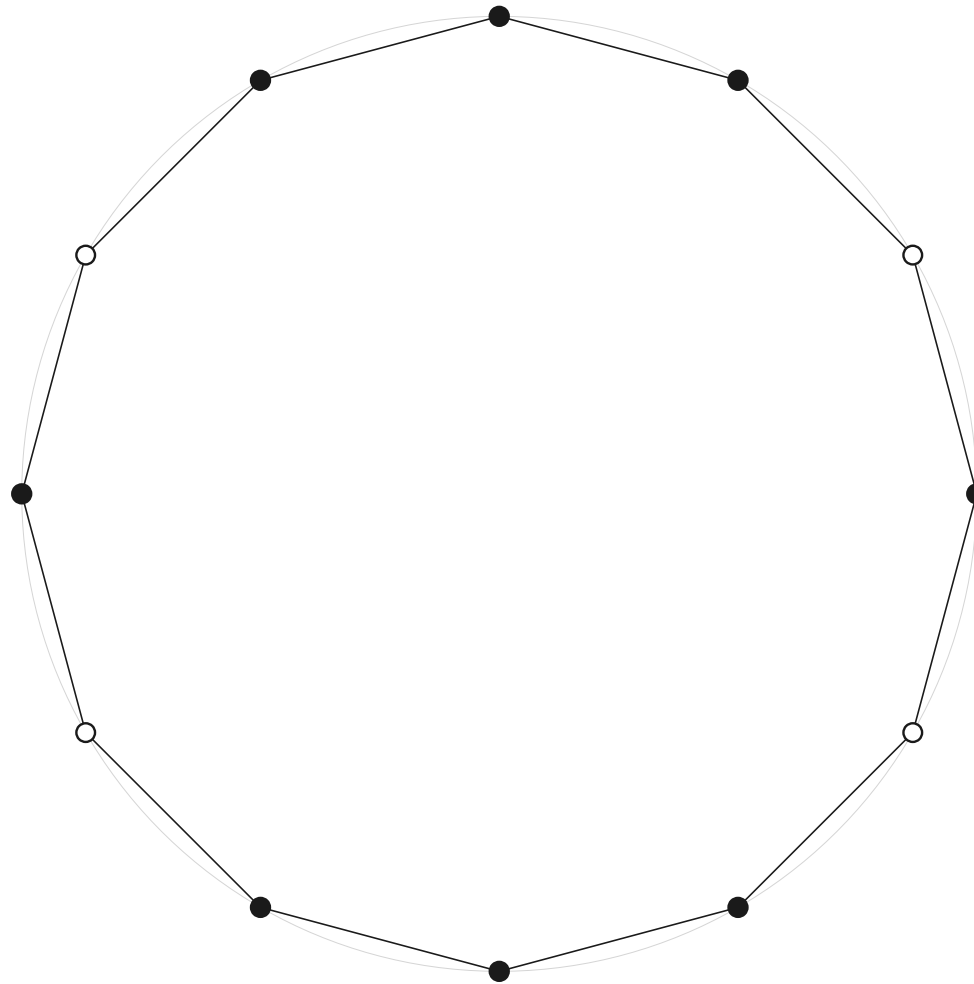
D3 $2 \cos(\pi/5) = \text{Phi}$, exactly -- not approached, metDetail · the marriage, $n=5$ 

$$\begin{array}{lcl} 2 \cos(\pi/5) & = & 1.6180339887 \\ \text{Phi} & = & 1.6180339887 \\ \text{diff} & = & 0.00e+00 \end{array}$$

D4

filled = shared with a smaller d-gon, hollow = new to n

Detail · vertex structure, n=12



4 of 12 vertices are new to n=12

reducible_count(n) vs (n-1) - totient(n)

checked, n = 2..60: 59 rows, 0 mismatches

n	reducible	totient predicts
5	0	0
6	3	3
10	5	5
12	7	7
17	0	0
24	15	15
30	21	21

